Contents

CM2013 — Signal Processing and Data Analytics in Biomedical Engineering 1
1. Course information . . . 1
2. Expected effort . . . 1
3. Contacts . . . 2
4. Intended learning outcomes . . . 2
5. Course structure . . . 3
6. Schedule (from TimeEdit, `TimeEdit_CM2013_2026-08-13_13_17.ics`) . . . 4
7. Examination . . . 5
8. Course literature . . . 9
9. Labs . . . 10
10. Canvas . . . 10
11. Ethical approach . . . 10
12. Generative AI policy . . . 10

CM2013 — Signal Processing and Data Analytics in Biomedical Engineering

Course Plan, Autumn 2026 (HT26)

Department of Biomedical Engineering and Health Systems, School of Engineering Sciences in Chemistry, Biotechnology, and Health, KTH Royal Institute of Technology.

Built from the legally-binding syllabus (`CM2013-20221 Syllabus.pdf`, Spring 22 edition 2), the previous Course PM (2022), and the VT25 Canvas offering.

1. Course information

- **Credits:** 7.5 HP
- **Grading scale:** A, B, C, D, E, FX, F (second cycle)
- **Main field of study:** Medical Engineering, Technology and Health

2. Expected effort

7.5 HP is a **200-hour** workload (KTH's standard 1 HP ≈ 26.7h). The estimate below is component-by-component, not a top-down guess, so it's checkable and revisable as the course actually runs.

Component	Hours	Type
Lectures (6 × 2h)	12	contact
Chapter reading + podcast, ×16	44	async
Lab notebooks, ×16	36	mostly async

Component	Hours	Type
— of which: the 2 booked lab-room sessions (2026-09-18, 2026-10-08)	5	contact, **subset of the line above, not additional**
Module quizzes, ×16	8	async
Capstone project — team + individual (booked project sessions, async team work, critical-comparison essay, teamwork admin)	70	mixed, incl. 12h booked sessions
Showcase	4	contact
Written exam — sitting it	4	contact
Written exam — preparation & review	18	async
Total	**196**	
Target (7.5 HP × 26.7h)	**200**	

196h against a 200h target — a tight, realistic fit (2% slack). Total scheduled contact time across the whole course (lectures + lab-rooms + project sessions + showcase + exam) is 37h; the remaining ~159h is self-directed. Two components carry the most estimation uncertainty and are worth watching once the course actually runs:

- **The capstone project (70h)** is the line most likely to run over in practice — "leakage-safe validation on a real dataset, done properly" reliably takes longer than teams expect the first time.
- **Exam preparation & review (18h)** is sized proportionally to the exam's weight (40 of 100 points), not derived from anything structural.

With near-zero slack, there's little headroom left if the project runs long again — worth revisiting this estimate after the course's first offering.

3. Contacts

- Examiner: Mikael Forsman, miforsm@kth.se
- Course responsible & teacher: Farhad Abtahi, sabt@kth.se
- Teacher, instructor for project and labs: Eduardo Illueca Fernández, eduardo.illueca@ki.se

4. Intended learning outcomes

From the legally-binding syllabus. After the course, students should be able to:

- **L1.** Describe and discuss the principles of biomedical signal acquisition, sampling, and processing
- **L2.** Characterize biosignals origin and noise features
- **L3.** Design and implement fundamental biosignal analysis, modelling, and visualization tools
- **L4.** Select and apply appropriate methods for pattern recognition and classification of biosignals for solving a given problem in biomedical engineering
- **L5.** Design, motivate, implement, and evaluate a signal processing method for solving a specific problem in biomedical engineering
- **L6.** Effectively work as part of a project team

5. Course structure

The course runs as **four thematic modules**, flipped-classroom: each module's book chapters, podcasts, and lab notebooks are worked through asynchronously; the scheduled lecture slots are synthesis and discussion, not first exposure to the material. Full chapter/asset detail: `module_map.md`.

Module	Theme	Chapters	Learning outcomes
M1	Biosignals Fundamentals & Acquisition	1–2	L1, L2
M2	Transforms, Spectral Analysis & Filtering	3–10	L1, L2, L3
M3	Machine Learning & Data Analytics	11–15	L3, L4, L5
M4	Project / Capstone	16 + capstone tracks	L5, L6

Team formation. Project teams (L6) are deliberately built to be multidisciplinary, not self-selected. The "Introduce yourself" discussion in the orientation module (run in week 1, before M1's lecture) asks each student for their degree background, prior signal/programming experience, and any clinical/domain exposure; the course team uses those answers to assign teams ahead of the M4 kickoff so each group mixes backgrounds rather than clustering by existing friend groups or single-discipline strength.

Group size. Expected cohort: 35–45 students. Teams of **3 by default, 2 where the numbers require it** — for 35–45 students that's roughly **12–15 teams**. Prefer 3s; use 2s only to make the count work, not as a first choice.

Track selection — 3 comparable tracks by default. Rather than spreading the cohort across all 6 capstone tracks (too thin per track at 12–15 teams, not enough for a meaningful shadow comparison), the default assignment pool is **Sleep-EDF, ECG (CinC2017), and EMG (Ninapro)** — 4–5 teams per track. These three land closest together on the *honest* baseline metric (Cohen's κ 0.52–0.84 on the reported eval mode), not just the nominal ★★★ label. **CTG (CTU-UHB) is not offered this year** — its honest baseline (κ ≈ 0.11) sits far below the other three, closer to a floor than a moderate challenge, so it was dropped rather than kept as an option. **HAR (★★, easier) is also not offered. BCI (★★★★, hardest)** remains available as the sole **opt-in stretch track**: a team can request it explicitly, instructor sign-off required, understanding the difficulty trade-off going in (its honest baseline barely beats chance — that's the point of the track, not a flaw).

Track assignment mechanism. Each team ranks its top 2 of the 3 default tracks via a short Canvas survey, submitted before the first project session (2026-09-29). The course team assigns one track per team, balancing preferences against keeping every track's team count roughly even (4–5 each). Assignments are published before 2026-09-29 so Iteration 1 (see `Project_Management_Guide.md`) starts on time.

Shadow pairing. At the same time track assignments are published, every team is paired with exactly **one other team on a different track** — one pairing per team, used for the rest of the course, not re-paired later. Purpose: every student individually writes a critical comparison of their own project against their shadow partner's (see `INDIVIDUAL_ASSESSMENT.md`'s critical-comparison essay, 5 of the individual 10 points) — comparing across genuinely different signals/tasks is the point, so pairing is deliberately **cross-track**, never two teams on the same track. Students should look in on their shadow partner's progress informally around the project's mid-point; the graded essay itself is written after the showcase, once they've seen the partner team's finished project.

6. Schedule (from TimeEdit, `TimeEdit_CM2013_2026-08-13_13_17.ics`)

Date	Time	Type	Location	Module
2026-08-31	10:00–12:00	Lecture	T4 (Curiesalen)	M1
2026-09-04	10:00–12:00	Lecture	T64 (Nätverkslabb)	M2
2026-09-18	13:00–16:00	Computer lab	T64 (Nätverkslabb)	M2
2026-09-21	10:00–12:00	Lecture	T54 (Elmqvist-salen)	M2
2026-09-29	10:00–12:00	Project	T63 (Theorellsalen)	M2
2026-10-08	13:00–15:00	Computer lab	T64 (Nätverkslabb)	M3
2026-10-15	14:00–16:00	Project	T63 (Theorellsalen)	M3
2026-10-23	14:00–17:00	Lecture	T63 (Theorellsalen)	M2/M3
2026-10-26	10:00–12:00	Project	T54 (Elmqvist-salen)	M3
2026-11-02	13:00–15:00	Lecture	T54 (Elmqvist-salen)	M3
2026-11-10	13:00–15:00	Project	T55 (Bråne-marksalen)	M4
2026-11-18	13:00–15:00	Lecture	T54 (Elmqvist-salen)	M3/M4
2026-11-24	10:00–12:00	Project	T63 (Theorellsalen)	M4
2026-12-01	10:00–12:00	Project	T63 (Theorellsalen)	M4

Date	Time	Type	Location	Module
2026-12-10	08:00–12:00	Presentation / showcase	T53 (Hertzsalen)	M4
2027-01-08	14:00–18:00	Written exam	T63 / T64	—

7. Examination

Per the legally-binding syllabus, examination is in three parts:

- **RED1 — Assignments, 2.0 credits, P/F.** Module quizzes across all 16 chapters. Each chapter's quiz draws **5 questions at random from a bank of 15** (so no two students necessarily see the same set), open-ended/short-answer, marked by hand rather than auto-graded. **Up to 3 attempts, 10 minutes per attempt, average of your attempts counts** — you can retake a quiz to improve your score, but each attempt is timed and the earlier ones still count toward the average, so treat every attempt as a real one rather than a scouting run.
- **PRO1 — Project work, 2.0 credits, P/F.** Group capstone project (one of the four default domain tracks in `companion-notebooks/tracks/`, HAR/BCI available on request) plus the individual critical-comparison essay and teamwork component.
- **TEN1 — Written exam, 3.5 credits, A–F.** Per the syllabus: *"The exam (A-F) determines the final grade for the course when all course parts have been passed. The examination form and grading criteria will be specified in a course-PM."* This PM specifies that criterion: the exam's A–F band is computed from the raw written exam score topped up by RED1/PRO1 point performance, as below.

Register for the exam yourself — this is not automatic. You must register for TEN1 (2027-01-08) through Ladok/KTH's exam registration system ahead of the deadline (check the current KTH-wide registration window — typically a few weeks before the exam date). This is a general KTH requirement, not specific to this course, but it's easy to miss if you're new to KTH's systems — many of you are starting your master's this term. Missing the registration window means missing the exam, regardless of how the rest of the course has gone. Set a reminder now, don't leave it until January.

A student must pass RED1 and PRO1 (complete the required deliverables) before TEN1's points determine a grade. Passing RED1/PRO1 is a gate, not a numeric contribution in its own right beyond the points shown below.

The written exam is mandatory — sitting it is not optional, even if a student's RED1+PRO1 points alone would already clear 51 (the F/E boundary). TEN1 is its own examination moment (3.5 of the course's 7.5 credits) with its own pass requirement; a point total is not a substitute for passing it. Practically, this matters because **students will not know their exact RED1/PRO1 point total before the exam date** (2027-01-08) — the showcase is only a month earlier (2026-12-10), and marking module quizzes, the project report, and the individual assessments takes longer than that. No student should walk into TEN1 assuming they've already "banked" enough points to coast — treat the exam as consequential regardless of how the term has gone.

Point system (TEN1 grade computation)

Component	Points	Exam moment
Module quizzes (16 chapters)	20	RED1
Project — team (code, report, presentation, deadlines)	30	PRO1
Project — individual (critical-comparison essay + teamwork)	10	PRO1
Written exam	40	TEN1
Total	**100**	

Full project rubric detail: `companion-notebooks/tracks/CAPSTONE_REPORT_RUBRIC.md` (team, 30 pts) and `companion-notebooks/tracks/INDIVIDUAL_ASSESSMENT.md` (individual, 10 pts).

Project rubric — how the 30 + 10 points break down

Seven team criteria are identical across all tracks; two are domain-specific (concretized per track). Every criterion is scored against a published 4-tier descriptor — this table shows what full marks looks like; the full tier-by-tier language (partial/minimal/missing) is in `CAPSTONE_REPORT_RUBRIC.md`.

#	Criterion	Points	Instrument	Full marks looks like
1	Pipeline integrity	3	Code	All seven pipeline stages separable and reusable
2	Signal processing rigor	5	Code (3) + Report (2)	Real DSP in the code, justified in the report; features constructed, not handed raw to the classifier

#	Criterion	Points	Instrument	Full marks looks like
3	Validation honesty	6	Code (3) + Report (2) + Presentation (1)	Correct split unit everywhere in code; full metric panel reported; leakage risk explained live
4	Reproducibility & engineering	2	Code	Seeded, config-driven, pinned environment, `smoke()` passes
5	Report quality & defence	3	Presentation	Design decisions justified against the spec; results read against a real ceiling; honest limitations
6	Deadlines	1	Process	Everything delivered on time, no extension needed

#	Criterion	Points	Instrument	Full marks looks like
7	Improvement over baseline *(domain-specific)*	4	Report	A controlled, evidence-based comparison against the supplied baseline — ≥2 alternatives tried one at a time, at least one falsified, and the final pipeline justified either by a named gain or by a defended decision to keep a lower/equal-scoring pipeline
8	Track-specific execution *(domain-specific)*	3	Report (2) + Presentation (1)	The specific leakage/imbalance/eval-mode risk for that track is named and handled
9	Iteration & revision history	3	`RESULTS.md` (2) + Presentation (1)	A dated iteration log with the metric **and its spread** per row, a decision log naming rejected alternatives, and **at least one decision explicitly revisited and changed** because of a later result

Team total	**30**

Soft gate: code failing Criterion 3's bottom row (no group-aware split) should not score above ~15/30 regardless of the rest — a leaked number is not evidence of anything.

Component	Points	What it checks
Critical-comparison essay	5	Individual, written, 300–500 words, due after the showcase — a specific comparison against your shadow-paired (different-track) team's project
Contribution & workload	2	Delivered your owned module(s), on schedule, at a quality peers relied on
Communication & collaboration	2	Proactive, responsive, attended check-ins
Conflict management	1	A disagreement or imbalance handled constructively, corroborated by peers
Individual total	**10**	

Teamwork is graded from three sources combined by the instructor — the team's ClickUp workspace (task log), a private peer-evaluation form, and a short individual reflection — never from one team member's self-report alone.

Grading scale

Points	Grade
91–100	A
81–90	B
71–80	C
61–70	D
51–60	E
below 51, near the boundary	FX (examiner's discretion — complementary assessment)
below 51	F

8. Course literature

- *Biomedical Signal Processing and Data Analytics: From Physiology to Machine Learning* — the course textbook (`Book_2026-07-28/`), 16 chapters + appendices.

- Chapter podcasts (one per chapter) and lecture slide decks — linked per module in Canvas; production checklist in `module_map.md`.

9. Labs

Every chapter has two notebooks, done in order, before that chapter's lecture: a **demonstration notebook** (reproduces the book's own figures on real data) and a **lab notebook** (a hands-on exercise with gaps you fill in). Both run in Colab. **Labs carry no separate points and there is no report to submit** — they're formative, and what the module quizzes and the capstone project both assume you've already practiced.

Two rooms are booked with a TA for drop-in support (2026-09-18 and 2026-10-08, both T64) — help with whichever lab is currently giving trouble, not "the" time to do all 16. Full detail, session-by-session: `Lab_Plan.md` (instructor-facing) and `Lab_Instructions.md` (student-facing); a short intro deck (`Lab_Intro.pptx`) covers the same ground for a first-lecture walkthrough.

10. Canvas

All course material, quizzes, and submissions run through Canvas. Information architecture (module/page tree): see sub-project B (`Canvas_IA.md`, to follow). Each module quiz is built in Canvas as a random-draw question group (5 of 15 questions per chapter, §7) with 3 attempts and a 10-minute timer per attempt; Canvas averages the attempts automatically.

11. Ethical approach

From the syllabus, verbatim:

- All members of a group are responsible for the group's work.
- In any assessment, every student shall honestly disclose any help received and sources used.
- In an oral assessment, every student shall be able to present and answer questions about the entire assignment and solution.

12. Generative AI policy

Stance: **"Allowed according to given guidelines."** Encouraged for idea generation, learning support, and troubleshooting code/algorithms — including your capstone project's code. Not permitted for writing the report, answering module quizzes, or the individual critical-comparison essay/teamwork reflection (English proofreading of your own report text is the one exception in the report). Full policy, disclosure format, and rationale: `Generative_AI_Policy.md` — written to paste directly into the course's own Canvas page, which already carries this exact template, unpublished.